Action-Conditioned Latent World Models for Synthetic Deformable-Object Dynamics

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Abstract

We study whether stiffness, elasticity, and deformability under contact are linearly readable from an action-conditioned latent world model of synthetic objects — the workshop topic *Physical Characteristics*. Observations come from an in-repo MLS-MPM renderer (2000 episodes, T=16, 64×64), not from captured video, and we do not claim results on PhysWorld or PhysTwin. The contribution is a controlled comparison: VICReg versus covariance-ablated / MSE-only / pixel-space alternatives, plus an action-conditioning knockout, against a particle-GNN baseline in the *same* mass-spring simulator. On a scaled physics dump (n_{val}=200), contact-quadrant linear probes beat chance (0.574 vs. 0.250) while stiffness class remains at chance (0.339 vs. 0.333), including an oracle probe on the EMA target. We report that limitation rather than a new SOTA number. Workshop ablations are a **2-seed range** (seeds 42, 43), not a mean $\pm$ std over a full multi-seed grid.

1 Introduction

Learned world models that forecast how a deformable object will respond to contact would be useful for manipulation, but the recent digital-twin literature is dominated by systems that start from *real video* and an MPM-optimized twin [Jiang et al., 2025, Yang et al., 2025]. This paper asks a narrower question on a fully synthetic stack: if we train an action-conditioned JEPA [Assran et al., 2023, Bardes et al., 2024] with VICReg [Bardes et al., 2022] on rendered MLS-MPM frames [Hu et al., 2018], which pieces of the objective and which form of action fusion actually matter, and what physical properties are linearly readable from the latent?

We do **not** claim to outperform PhysWorld or PhysTwin. Those methods operate in a different domain (real captures, reconstructed geometry, privileged particle twins). Our numbers are within-simulator. The empirical payload is (i) a six-way baseline on a shared mass-spring split (Table 1), (ii) a 2-seed VICReg vs MSE comparison on the physics dump (Table 2; range, not a std), and (iii) probe diagnostics, including an effective-rank participation ratio on z_{target} (Section 5).

2 Related work

JEPA and VICReg. I-JEPA [Assran et al., 2023] and V-JEPA [Bardes et al., 2024] learn by predicting embeddings of a target view from a context view, with an EMA target encoder in the BYOL family [Grill et al., 2020]. AC-JEPA instantiates that recipe on $(x_t, a_t) \mapsto z_{t+1}$. Collapse is regularized with VICReg [Bardes et al., 2022]: invariance, a per-dimension variance hinge, and an off-diagonal covariance penalty, implemented in `src/losses.py`.

Particle-graph world models. DPI-Net [Li et al., 2019] and GNS [Sanchez-Gonzalez et al., 2020] propagate messages on particle graphs to predict the next state. Our particle-GNN baseline reuses

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the simulator's spring `_edge_list` and is scored in graph-readout space; it is a within-simulator relative of that family, not a reimplementation of those papers' datasets.

Real-video digital twins (domain gap). PhysTwin [Jiang et al., 2025] reconstructs and resimulates deformables from sparse video with a physics-informed twin. PhysWorld [Yang et al., 2025] fits an MPM twin to real video, synthesizes demonstrations, and trains a GNN for real-time prediction. Both start from captured footage of a small number of objects. Our dump is synthetic RGB of disks/bars with three global Young's moduli, no camera, no residual physics mismatch. Table 1 is therefore not a SOTA table against those systems.

3 Method

Architecture. Following `src/models/ac_jepa.py` and `ModelConfig`, AC-JEPA has three modules. An online encoder f_θ (ViT by default) maps a frame x_t to z_t. An action-conditioned predictor g_ϕ maps (z_t, a_t) to $\hat{z}_{t+1}$. A target encoder $f_{\bar{\theta}}$ is an EMA copy $\bar{\theta} \leftarrow \tau\bar{\theta} + (1-\tau)\theta$ with default τ=0.996; it produces the stop-gradient target $z_{t+1} = f_{\bar{\theta}}(x_{t+1})$. Only f_θ and g_ϕ receive gradients. Workshop YAML uses 64×64 frames, patch size 8, embedding width 128, concat fusion, and $a_t \in \mathbb{R}^4$ (contact xy + force xy). Setting `condition_on_action=False` drops a_t so g_ϕ sees z_t only (no-action ablation).

Objectives. `prediction_objective` in `src/losses.py` implements four latent losses on $(\hat{z}_{t+1}, z_{t+1})$: (i) full VICReg (λ=25, μ=25, ν=1, γ=1); (ii) Variant A, `vicreg_no_cov`, invariance + variance only; (iii) Variant B, invariance MSE only; (iv) optional smooth-ℓ_1 / cosine stand-ins for the invariance term. Variant C is a pixel-space autoencoder that decodes $g_\phi(z_t, a_t)$ to $\hat{x}_{t+1}$ rather than matching z_{t+1}. Open-loop evaluation applies the predictor for k steps without replacing $\hat{z}$ by the true encoding.

4 Experiments

Data. Prompt 1 writes `outputs/physics_mpm/`: 2000 episodes of length 16, 64×64 RGB, backend `numpy` MLS-MPM (not a Taichi install), dump seed 0, geometries disk/bar, $E \in \{18, 70, 220\}$ with counts $667/667/666$ (`outputs/dataset_report.json`). The workshop split is n_{train}=1800, n_{val}=200. Prompt 4 baselines instead use an in-process mass-spring lattice (n_{train}=120, n_{val}=40, 8 epochs, device `cpu`).

Within-simulator baselines. Table 1 is generated from `outputs/baseline_comparison.csv` (no hand-copied cells). VICReg latent MSE at k=1, 5, 10 is 0.145, 0.174, 0.181. Dropping the action (`no_action`) raises k=5 MSE from 0.174 to 0.462 on this split: the predictor uses a_t when it is available. MSE-only records a much smaller latent MSE ($5.20e-04$ at k=1) — as expected when the objective is pure invariance and is free to collapse directions VICReg would keep alive. The particle GNN's k=1 MSE ($3.32e-08$) is *not* in the same space; we do not rank it against ViT latents. Stiffness probes on this split sit near chance for every method (e.g. VICReg 32.5% vs. 33.3%).

Loss ablation (2-seed range). Table 2 is generated from `outputs/ablation_twoseed.json`. Cells are mean $\pm$ half-range over seeds 42 and 43, **not** a sample std. We did not run the originally planned multi-seed variant grid. Dummy 1-epoch lattice numbers are not used as workshop results.

Qualitative rollouts. Figure 1 shows held-out MLS-MPM episodes at $k \in \{1, 5, 10\}$ for $E \in \{18, 70, 220\}$. VICReg columns use a *post-hoc* frozen-latent pixel decoder that is not part of AC-JEPA; the Pixel-AE row is a short qualitative fit unless a finished `outputs/pixel_ae/` checkpoint exists (`outputs/figures/README.txt`). Figure 2 is validation episode 1841, the worst latent MSE at k=10 (0.9651; n_{val}=200).

Table 1: Within-simulator baseline comparison on the mass-spring lattice ($n_{\text{train}} = 120$, $n_{\text{val}} = 40$, 8 epochs, device `cpu`). Open-loop rollout MSE (↓) at $k \in \{1, 5, 10\}$. Linear-probe accuracy (↑, %) at k=1; chance is 33.3% stiffness / 25.0% contact. Particle-GNN `rollout_mse` is in pooled graph-readout space, not the ViT latent — magnitudes are not interchangeable across rows. Source: `outputs/baseline_comparison.csv`.

Method	k=1	k=5	k=10	Stiff. %	Contact %
AC-JEPA (VICReg)	0.145	0.174	0.181	32.5	40.0
AC-JEPA (no covariance)	0.247	0.257	0.273	30.0	32.5
AC-JEPA (MSE-only)	**5.20e-04**	**0.021**	**0.022**	40.0	32.5
Pixel autoencoder	0.335	0.325	0.326	40.0	32.5
No action conditioning	0.239	0.462	0.362	40.0	32.5
Particle GNN (GNS/DPI-Net family)	3.32e-08	8.41e-08	9.81e-08	40.0	32.5

Table 2: Workshop VICReg vs MSE on the MLS-MPM dump, two seeds (42, 43). Table pending `outputs/ablation_twoseed.json`.

Two-seed range not written yet.

5 Limitations

Probes. Smoke-split probes in `outputs/metrics.json` use n_{val}=12 and are underpowered (binomial SE ≈ 0.14 at chance 0.333). On the physics dump we do have a powered readout. Findings copied from `outputs/probe_diagnostics.json`:

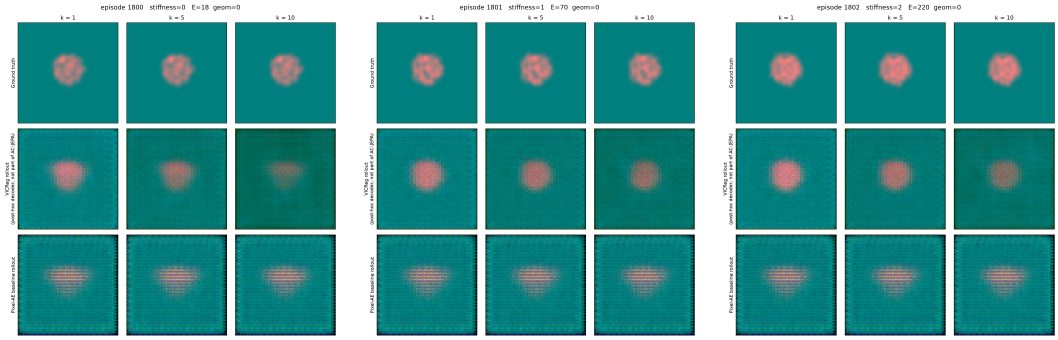
> Linear-probe findings copied from `outputs/probe_diagnostics.json` (`models.vicreg_physics_mpm.findings`):
>
> - stiffness class: after scaling to n_val=200, mean val acc=0.339 is still consistent with chance=0.333 (z=0.16, SE≈0.033). This is a real limitation: a linear readout of stiffness class from the latents does not beat chance.
> - contact quadrant: mean val acc=0.574 is above chance=0.250 (z=10.58, n_val=200).
> - stiffness class (z_target): after scaling to n_val=200, mean val acc=0.331 is still consistent with chance=0.333 (z=-0.08, SE≈0.033). This is a real limitation: a linear readout of stiffness class (z_target) from the latents does not beat chance.
>
> Checkpoint caveat (same file, `checkpoint_note`): Copied from outputs/vicreg/best.pt during the in-progress 200-epoch workshop run (not the finished checkpoint).; Not evaluated: workshop MSE run has not started yet.

Mean stiffness-class val acc is 0.339 vs. chance 0.333; the EMA-target oracle is 0.331; train acc is 0.350; Young's-modulus regression R^2 averages -0.002. Contact-quadrant val acc is 0.574 vs. chance 0.250. z_{target} occupies a near-constant shell ($\|z\|$ std 1.1e-04; 128/128 dims below the VICReg γ=1 hinge; hinge binding `true`). We treat stiffness non-decodability as a limitation of this encoder, not a result to suppress. Probe numbers are from epoch 133 of 200 (git `07167b5`).

Effective rank. Effective-rank participation ratios will be filled from the final 200-epoch probes (`effective_rank` in `outputs/probe_diagnostics.json`).

2-seed ablation, not a multi-seed study. Table 2 uses 2 seeds. Half-range is not a confidence interval. The originally planned multi-seed variant grid was skipped for the archival deadline.

Domain gap and scale. PhysWorld and PhysTwin start from real video and a reconstructed MPM twin. We render 2000 short (T=16) synthetic clips at 64×64 with 3 discrete global E values. That is a different data regime in appearance, geometry, and parameter identification. Nothing here should be quoted as evidence on those benchmarks.



E=18 (ep. 1800) E=70 (ep. 1801) E=220 (ep. 1802)

Figure 1: GT | post-hoc-decoded VICReg | Pixel-AE at $k \in \{1, 5, 10\}$ on the MLS-MPM dump. VICReg weights are **not** finished (133 of 200 epochs). Source: `outputs/figures/`.

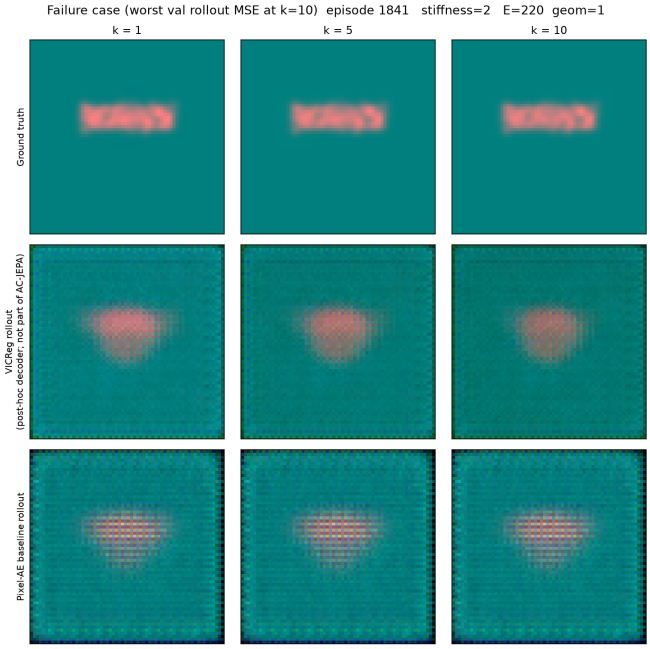


Figure 2: Episode 1841 (E=220), worst val latent MSE at k=10 (0.9651). Caption text is `outputs/figures/failure_case_caption.txt`, grounded in `probe_diagnostics.json`: stiffness stays at chance, z_{target} is collapsed, contact probes beat chance. Not a contact-extrapolation story.

6 Conclusion

AC-JEPA on synthetic deformables is a setting in which action conditioning changes latent rollout error on a shared mass-spring split, VICReg's extra terms do not automatically win latent MSE against a pure invariance baseline, and a scaled linear probe can read contact quadrant but not stiffness. Those are the claims this repository currently supports. They are not a SOTA result on real video.

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